

16.485 - Visual Navigation for Autonomous Vehicles (VNAV) Syllabus



[picture courtesy of [Skydio](#)]

Lectures and Labs:

- *Lectures*: Monday/Wednesday/Friday 1-2pm in 35-225
- *Labs*: Wednesday 3-5pm in 33-116
- *Lab handouts*: <https://vnav.mit.edu/>

Websites:

- *Canvas*: <https://tinyurl.com/VNAV2024-canvas> (announcements, material)
- *Piazza*: <https://tinyurl.com/VNAV2024-piazza> (Q&A with TAs and Instructors)
- *Gradescope*: <https://tinyurl.com/VNAV2024-gradescope>
- *Link to this syllabus*: <https://tinyurl.com/VNAV2024-syllabus>
- *Admission survey*: <https://tinyurl.com/VNAV2024-survey>

Class Staff:

- *Instructors*: Luca Carlone
- *Teaching Assistants*: Aaron Ray, Andrew Fishberg

Other Info:

1. *Prerequisites:* real-time systems programming (16.35 or similar - **we are going to code in C++**), optimal estimation and control (16.32 or similar), linear algebra (18.06 or similar), or permission of the instructors. Assignments are largely based on ROS and OpenCV and are all in C++. Previous experience with robotics (16.405 or similar) is a plus.
C++ refresher: https://youtu.be/F_vIB3yixaM
2. *Units:* 3-2-7 (graduate-level subject)

Lectures and Laboratory Schedule:

Week	Dates	Lecture topic	Lab	Platform
1	Sep 4, 6	Introduction	Lab 1: Linux, C++, Git	-
2	Sep 9, 11, 13	3D Geometry	Lab 2: ROS	-
3	Sep 16, 18	Geometric Control	Lab 3: 3D trajectory following	simulator
4	Sep 23, 25, 27	Trajectory Optimization	Lab 4: 3D trajectory optimization	simulator
5	Sep 30, Oct 2, 4	2D Computer Vision	Lab 5: feature detection	simulator
6	Oct 7, 9, 11	2-view Geometry and Minimal Solvers	Lab 6: object localization	simulator
7	Oct 16, 18	Non-minimal Solvers and Visual Odometry	Lab 7: GTSAM	simulator
8	Oct 21, 23, 25	Place Recognition	Lab 8: ML for robotics	simulator
9	Oct 28, 30, Nov 1	SLAM and Visual-Inertial Navigation	Lab 9: SLAM	simulator
10	Nov 4, 6, 8	Advanced Topics	Final project	depending on project
11	Nov 13, 15	Advanced Topics	Final project	depending on project
12	Nov 18, 20, 22	Advanced Topics	Final project	depending on project
13	Nov 25, 27	Guest Lectures	Final project	depending on project
	Thanksgiving Break			
14	Dec 2, 4, 6, 9, 11	Students Presentations	Final project	depending on project

Advanced Topics lectures will be decided depending on interest and guest lecturers.

Assignments and Grading:

Graded **assignments** include lab exercises and a final project. Lab exercises include a set of questions covering both theoretical aspects (discussed during the lectures) and the result of the implementation and testing of algorithms (assigned during the labs). The code written in some

of the labs will also be subject to evaluation. Assignments for Lab 1 and 2 are individual. Labs 3-9 include a mix of team assignments (we expect teams of 2 students) and individual assignments. Lab exercises are typically announced on Wednesdays and the corresponding assignments are due the following Wednesday. VNAV does not have a final exam.

The final project will be completed in teams, formed according to the students' interests. The grading for the final project will be based on:

- 1) Technical report (formatted according to ICRA guidelines).
- 2) Final demo, showcasing the outcome of the project.
- 3) Team presentation, including videos describing the project and its implementation.

The following is a list of important dates for the assignments:

Week	assignment	Individual/ Team	Announced	Due
1	Lab 1: Linux, C++, Git	I	W Sept 4	W Sept 11
2	Lab 2: ROS	I	W Sept 11	W Sept 18
3	Lab 3: 3D trajectory following	T/I	W Sept 18	W Sep 25
4	Lab 4: 3D trajectory optimization	T/I	W Sep 25	W Oct 2
5	Lab 5: Feature detection	T/I	W Oct 2	W Oct 9
6	Lab 6: Object localization	T/I	W Oct 9	W Oct 16
7	Lab 7: GTSAM	T/I	W Oct 16	W Oct 23
8	Lab 8: Place recognition	T/I	W Oct 23	W Oct 30
9	Lab 9: SLAM	T/I	W Oct 30	W Nov 6
10-14	Final project (report, demo, presentation)	T	W Nov 6	M Dec 2 - W Dec 11

The grading policy is as follows:

- **[65%] Lab exercises:** Set of questions covering both theoretical aspects (discussed during the lectures) and the result of implementation and testing of algorithms (assigned during the labs). You will upload your solutions and code produced during the labs to a github.mit.edu repository.
- **[25%] Final project report, demo, presentation:** A final demo, together with a technical report (formatted according to ICRA guidelines) detailing and motivating the choice of final project and your approach to address and solve the corresponding problem. A final presentation reporting on the outcome of your implementation, including experimental results and videos describing the overall project.
- **[5%] Participation, TA evaluation:** Your participation in the lectures and labs, including attendance and evaluation of your performance in the labs by the VNAV staff.
- **[5%] Team members' assessment:** Assessment of your performance by your teammates.

Resources:

There is **no required textbook**. The following are **recommended** books (also available online):

- "State Estimation for Robotics" by Timothy Barfoot, Cambridge University Press (ISBN: 9781107159396). *[This text is on reserve on MIT Libraries]*

- “*An Invitation to 3-D Vision: From Images to Geometric Models*” by Yi Ma, Jana Kosecká, Jana, Shankar Sastry, and Stefano Soatto, Springer (ISBN: 9780387008936). *[This text is on reserve on MIT Libraries]*

Pointers to other relevant references (tutorial/survey papers as well as technical papers) will be provided during each lecture. VNAV heavily relies on our lecture notes (available on Canvas and [here](#)). **It is recommended to skim/read the lecture notes before the corresponding lecture to maximize your understanding of the material.**

Additional Policies:

- **Late work** will receive reduced credit, according to the following policy: if the grade achieved by a submission is X , the grade resulting from a late submission is computed as: $X_{\text{delayed}} = X * \max(1 - 0.15 * d, 0)$, where d is the number of days of delay. For instance, if $X = 10$ and there is no late submission, then $X_{\text{delayed}} = 10$, but if the assignment is submitted 2 days after the deadline, then $X_{\text{delayed}} = 10 * (1 - 0.3) = 7$.
- **Collaboration is essential** in the field, and therefore is essential for all assignments. Within teams, teamwork is an absolute necessity, and we expect that teams will work together to generate the technical content and solve the exercises for each lab.
- Across teams, we encourage collaboration and discussion. However, we explicitly forbid the appropriation of code, data, plots, or writing across teams, even with modifications or paraphrasing. In other words, **any writing included in a lab assignment must be authored by your team**; any data or plots must come from your team. You must also explicitly credit any collaborators.
- For the **final project**, full collaboration within the team on all aspects of the challenge is encouraged. Every member of the team will be expected to contribute a roughly equal share to the design, implementation, and presentation of the challenge.

Please contact a member of the course staff for any clarification about the policies above.

Extra resources to improve your communication skills. The AeroAstro Comm Lab provides field-specific 1-1 peer coaching sessions for the AeroAstro community. Our specially trained Comm Lab Fellows can help you enhance your papers, presentations, posters, or any other communication task. For more information on booking a free appointment, please visit <https://mitcommlab.mit.edu/aeroastro/> or in-person in 33-320.

Use of Generative AI. You may use Generative AI tools, such as ChatGPT or similar, to help generate ideas and brainstorm. However, you should note that the material generated by these tools may be inaccurate, incomplete, or otherwise problematic. Beware that using these tools may also stifle your own independent thinking and creativity.

If you use these tools in VNAV, the instructor encourages you to share interesting outcomes (e.g., surprising solutions or interesting failure modes). You will not be penalized in any way for using these tools, but at the same time, you are required to disclose their use and reference precisely which aspect of the assignment is AI generated and which tool has been used.